THE PRESHEAF COLIMIT CONSTRUCTION AGREES WITH F_! FOR SEPARATED LOCALLY QUASI-FINITE MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of f_! for separated morphisms (Definition definition-f-shriek-separated, canonical tag 0F4Z). cited at statement lines 1126-1128. The sheaf whose agreement with the presheaf construction is claimed.
- Colimit defining the presheaf f_{p!} (equation-colimit-definition, canonical tag 0F6M). cited at proof lines 1137-1139. The colimit that becomes directed in the separated locally quasi-finite case.

Source reference. Original definition 0F4Z: sources/stacks-chow/tags.txt, source lines 18919–18919. Cited at statement lines 1126-1128: the sheaf whose agreement with the presheaf construction is claimed.

Source reference. Original equation 0F6M: sources/stacks-chow/tags.txt, source lines 18977–18977. Cited at proof lines 1137-1139: the colimit that becomes directed in the separated locally quasi-finite case.

Lemma 0.2. *Let $f : X \to Y$ be a separated and locally quasi-finite morphism of schemes. Functorially in $\mathcal{F} \in Ab(X_{étale})$ there is a canonical isomorphism(!)*

$$f_{p!}\mathcal{F} \longrightarrow f_!\mathcal{F}$$

of abelian presheaves which identifies the sheaf $f_!\mathcal{F}$ of Definition 0F4Z with the presheaf $f_{p!}\mathcal{F}$ constructed above.

Proof. Let V be an object of $Y_{étale}$. If $Z \subset X_V$ is locally closed and finite over V, then, since f is separated, we see that the morphism $Z \to X_V$ is a closed immersion. Moreover, if Z_i, $i = 1, \ldots, n$ are closed subschemes of X_V finite over V, then $Z_1 \cup \ldots \cup Z_n$ (scheme theoretic union) is a closed subscheme finite over V. Hence in this case the colimit (0F6M) defining $f_{p!}\mathcal{F}(V)$ is directed and we find that $f_{!p}\mathcal{F}(V)$ is simply equal to the set of sections of $\mathcal{F}(X_V)$ whose support is finite over V. Since any closed subset of X_V which is proper over V is actually finite over V (as f is locally quasi-finite) we conclude that this is equal to $f_!\mathcal{F}(V)$ by its very definition. □

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